PERGAMON

Journal of Transport Geography 9 (2001) 39–48

JOURNAL OF TRANSPORT GEOGRAPHY

www.elsevier.com/locate/jtrangeo

Impacts of the National Trunk Highway System on accessibility in China

Si-ming Li *, Yi-man Shum

Department of Geography, Hong Kong Baptist University, Kowloon, Hong Kong

Abstract

Recently, China has launched a major programme of motorway construction – the National Trunk Highway System (NTHS). The present study analyses the impacts of this highway development programme on the pattern of accessibility gradients, trying to draw implications for regional growth. Obviously, this programme will bring about substantial improvements in accessibility across the nation. There is evidence, however, that highway investment exhibits diminishing returns over time. Greater improvements in the nodal accessibility of the major coastal cities, as compared with cities in interior and periphery provinces, in the initial stage of the highway development programme are found. But as time progresses, the NTHS will bring about more balanced development in the spatial sense.

Keywords: National Trunk Highway System; Valued graph; Accessibility; China

1. Introduction

China is currently undertaking a national motorway development project, the National Trunk Highway System (NTHS) (China Transport and Communications Yearbook, 1995, pp. 143–144; Gao, 1998). The NTHS consists of "five vertical (i.e., north–south) and seven horizontal (i.e., east–west)" core routes, with a total length of 35,000 km. [1] When completed, the NTHS will provide a network of high speed limited access road links to connect the national capital (Beijing) with all other directly administered municipalities (Tianjin, Shanghai and Chongqing), all provincial capitals, all other cities with population of one million or above and the majority of cities with population in excess of 500,000. The NTHS, in fact, forms part of a more extensive highway network system, the National Highway System (NHS), which comprises a total of 70 routes, covering almost every corner of the country. Some sections of the NHS outside the NTHS will also be upgraded to near motorway standard. The scale of the NTHS can only be matched by the Interstate Highway System of the United States launched in the 1950s and 1960s (Taaffe et al., 1996; Garrison, 1960), and perhaps also the trans-European road network launched in the 1990s, which is due for completion in 2002 (Gutiérrez and Urbano, 1996). The latter, however, is just an extension of already extensive motorway systems of the individual countries within the European Union.

The NTHS project exemplifies the accelerated shift from the railway to the road as the dominating form of land transport in China in recent times. Between 1978 and 1997, freight traffic by road, as measured in tonne-km, increased at an average rate of 19.8% per annum (SSB, 1998, pp. 539–540; figures quoted below are from the same source). Whereas the road traffic accounted for only 2.8% of the total freight movement in 1978, its share increased to 13.8% in 1997. At the same time, the railway's share decreased from 54.4% to 34.3%. In terms of passenger transport, the increase in road usage has been equally large. Between 1978 and 1997, road usage, as measured in passenger–km, increased from 52.1 billion to 554.1 billion, or at an average rate of 13.3% per annum. Road's share of passenger traffic also increased from 29.9% to 55.3%, whereas railway's share decreased from 62.7% to 35.4%. With the introduction of the NTHS, the shift to road transport will accelerate in the years to come.

* Corresponding author. Tel.: +852-2339-7121; fax: +852-2339-5990.

E-mail address: lisiming@hukbu.edu.hk (S. Li).

[1] The five "vertical" routes, from east to west with the exception of the first, are: (i) Tongjiang–Sanya (Hainan Island) (ii) Beijing–Fuzhou, (iii) Beijing–Zhuhai, (iv) Erlianhote–Hekou and (v) Chongqing–Chiangjiang. The seven "horizontal routes", from north to south, are: (i) Suifenhe–Manzhouli, (ii) Dandong–Lasha, (iii) Qingdao–Yinchuan, (iv) Lianyungang–Torrogort, (v) Shanghai–Chengdu, (vi) Shanghai–Ruili and (vii) Hengyang–Kunming.

0966-6923/01/$ - see front matter
PII: S0966-6923(00)00040-5

Implementation of the NTHS, naturally, has to be undertaken in stages. It was first put forward by the Ministry of Communications in the late 1980s. In fact, China began to build its first motorway, the Shenyang Dalian Expressway, as early as 1984. The early 1990s witnessed the beginning of China's massive drive to upgrade its highway network, with the gradual implementation of the NTHS. According to the time frame laid down by the Ministry of Communications, by the year 2000, 17,000 km of trunk highways, composed of two vertical routes (Tongjiang–Sanya and Beijing–Zhuhai) and two horizontal routes (Lianyunggang–Torrogrt and Shanghai–Chengdu), will have been completed. The entire system will be completed by the year 2020. Through various means, such as international bank loans, build-operate-transfer (BOT) arrangement, joint ventures with foreign capital, and local government investments, China, both at the national and local levels, has been able to raise substantial amounts of money for highway investment projects. For example, as of 31 August 1998, The Asian Development Bank and the World Bank had extended loans amounting to US$1.70 billion and US$4.02 billion to China, respectively, to assist construction of 1675 and 4229 km of motorways (ADB, 1998). Also many Hong Kong based conglomerates such as the Cheung Kong Group and the New World Group have set up subsidiaries specializing in infrastructure primarily highways investment in China, and many China based infrastructure development firms have sought listing in the Hong Kong Stock Exchange. As of the end of 1997, a total of 4771 km of motorways had been completed (SSB, 1998, p. 537). The majority of these motorways fall within the NTHS. Fig. 1 shows the NTHS, in conjunction with the NHS. Different symbols are employed to show: (i) sections in operation in 1997, (ii) the two vertical and two horizontal routes planned for completion in 2000, (iii) the rest of the NTHS up to 2020 and (iv) section of the NHS outside the NTHS.

The present paper attempts to analyse the changing accessibility gradients and their impacts on China's regional economy with the staged implementation of this huge project. Accessibility measures based on the graph theoretic approach will be employed (Taaffe et al., 1996). Despite the scale of the highway development project and its potential impacts on China's space economy, research on the topic has been scanty. A recent paper by Loo (1999) briefly touches upon the emerging motorway network in the Pearl River Delta Region, Guangdong Province. To our knowledge, however, to date there have been no published works on motorway development in China at the national scale. Below we first detail the procedures used to construct the accessibility measure. We then present the computation results.

2. "Valued graph" and the minimum distance matrix

2.1. Definition

A complex transport network may be abstracted as a graph in the topological sense, consisting of vertices

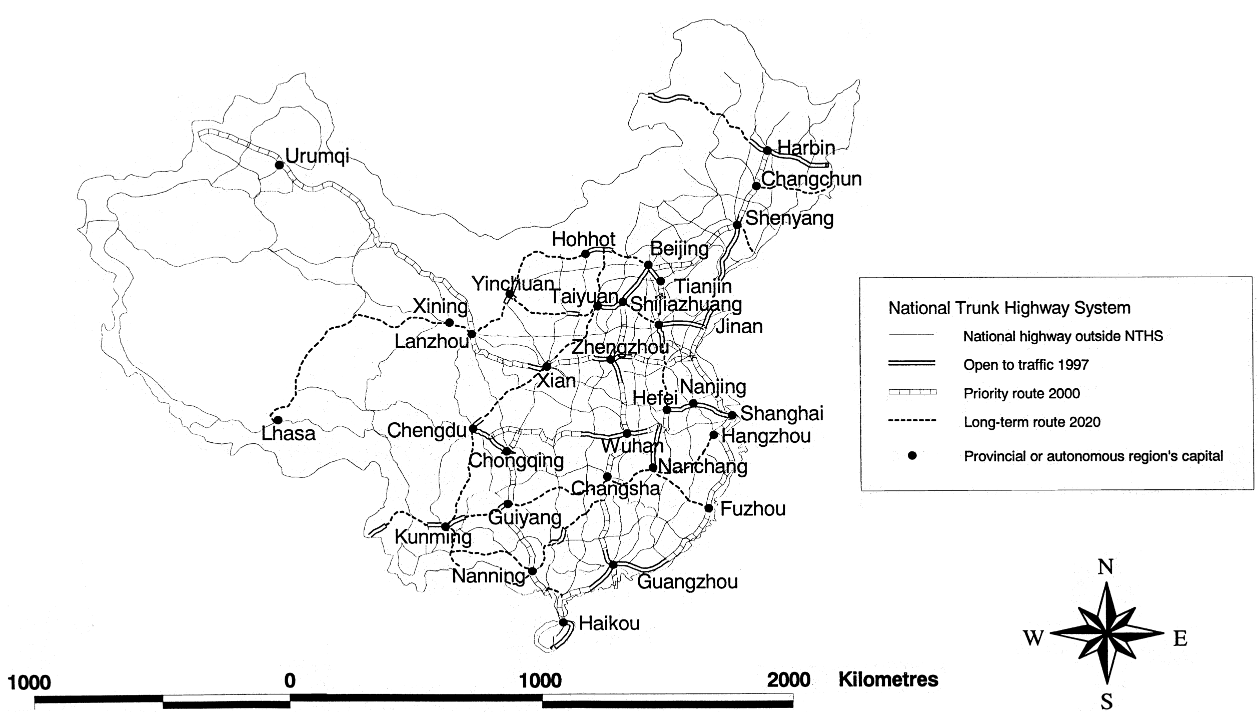


Fig. 1. National Trunk Highway System in China.

(nodes, or, in the present case, cities), and edges (links, i.e., highways) (Taaffe et al., 1996, pp. 249–286; Gabriel and Vaclav, 1996; Wheeler and O'Kelly, 1999). These sets of nodes and links can be represented in the form of a matrix with elements comprising two numbers, "0", indicating the absence of direct connection, and "1", indicating the presence of direct connection, between a pair of nodes. Let C denote this matrix, where the rows indicate origins and the columns indicate destinations. One measure to describe a network topologically is its *diameter*, which is defined to be the number of links needed to connect the two most remote nodes in the network. The matrix $D = C^n$, where n is the diameter of the network, is called the Shimbel distance or D-matrix. The elements of D give the minimum number of steps or links needed to be taken to connect the corresponding pairs of nodes. Consider row i in D. Summing its elements gives the minimum number of links required to connect i to all other nodes in the network. Let v_i be this sum. Then v_i is a measure of how accessible i is in respect to other nodes in the network.

The problem with the Shimbel D-matrix and hence v_i as an accessibility measure is that they take only into consideration the topological properties of the network and do not reflect the actual distance (physical or otherwise) needed to go from i to other places in the network. This may be adequate if the aim is primarily to describe how the various nodes in the network are connected to each other, as would be the case for a telecommunications network (Wheeler and O'Kelly, 1999) or how well one airport is linked to other airports (Ivy, 1995). However, for a transport network, there is an apparent need to incorporate measurements of distance in order to gauge the accessibility of different places. An alternate procedure is to construct a matrix L, the elements of which are:

(i) the physical or time distance (or any other distance measure) between a given pair of nodes i and j;

(ii) ∞ in case there is an absence of direct link between i and j.

L is then a matrix of a "valued graph" (Taaffe and Gauthier, 1973, pp. 138–148). Define the following "multiplication" operation:

$$\sum l_{ik} \cdot l_{kj} = \min\{(l_{ik} + l_{kj}),\ \text{all } k\}. \tag{1}$$

Then the matrix $S = L^n$, where n is the diameter of the network, is a matrix of minimum distance. Its typical element s_{ij} gives the minimum distance, whether measured in physical or time terms, to go from i to j. The row sum,

$$a_i = \sum s_{ij}, \tag{2}$$

gives the minimum distance, in physical or time terms, needed to travel between i and all other nodes in the network. As a measure of accessibility for a particular place i, or *nodal accessibility*, a_i contains more information than v_i does, and is therefore a better index if the aim is to describe more than the network's topological properties. The sum,

$$\sum a_i,$$

gives the minimum distance to travel from all nodes to all other nodes in the network. It is thus a measure of the accessibility of the entire network.

These minimum distance accessibility indices, a_i for a given node or place as given in (1), and $\sum a_i$ for the entire network, are employed in subsequent analysis. The construction of the NTHS primarily affects the time needed to travel from one place to another and not the actual physical distance traversed. Thus, only time distance is employed to measure the change in the accessibility gradients. Note that a similar approach has been adopted by Murayama (1994) in his study of railway development in Japan. Gutiérrez et al. (1996) study of the high-speed train network in Europe and Gutiérrez and Gómez (1999) work on the M-40 orbital motorway in Madrid also analyse the change in time distance as a result of transport infrastructure development. However, in these two studies a weighting factor measuring the "mass" of the individual centres has been added in the calculation of the accessibility score. On a priori grounds it is difficult to judge whether the incorporation of the mass term is a better measure or not. But what constitutes the mass term is always subject to debate. Should we use population? Or should we use the gross domestic product or other measures of the relative importance of the individual centres? In addition, there is the question of the relevant time frame (Linneker and Spence, 1996). Should we just use the base year mass figures? Or should we try to incorporate projections of the mass term over the forecasting period? Leaving out the mass term certainly circumvents these problems. Moreover, it tends to sharpen the pure location effects of time distance reductions. [2] This enables us to better grasp the accessibility changes of the more remote centres, which in the case of China would have been associated with much smaller weights had a mass term been employed. We could have used other accessibility measures, especially those based on the gravity model. But the accessibility scores obtained in gravity-type formulations do not lend themselves to ready interpretation. Also the impedance function employed has a built-in distance

[2] We would like to thank an anonymous referee for pointing out that the present approach is more locational than economic.

decay element, which tends to de-emphasize improvements in the more remote and isolated regions of the country (Bruinsma and Rietveld, 1998; Gutiérrez and Gómez, 1999).

2.2. Measuring changing time distance in NTHS

The NTHS is a long-term development plan. Only isolated sections of it have been completed to date. Hence it is at present hardly a network. But the different segments of the NTHS also constitute parts of the NHS, which is a fully interconnected network system. In fact, as indicated above, the entire NTHS, when completed as planned in the year 2020, still constitutes a portion of the NHS, although the NTHS itself will then be fully interconnected and forms a network of its own. In the present analysis, we examine how time distances change in this portion of the NHS, i.e., the portion that will become the NTHS, with the gradual implementation of the latter system.

The directly administered municipalities and provincial capitals occupy special positions in China's urban hierarchy (Tang, 1997; Zhang and Zhao, 1998). A major objective of the NTHS is to link up the country's 31 provincial capitals and directly administered municipalities. These cities thus constitute the nodes or vertices in the proposed NTHS. The edges or links are the segments linking pairs of nodes in the portion of the NHS that have been upgraded or will be upgraded to form the NTHS. It is assumed that travel speed averages 40 km/h for sections of the NHS that have not been upgraded to motorways, and 100 km/h for sections that have been upgraded and become part of the NTHS.

For the unimproved sections, 40 km/h may appear too low, as all highways in the NHS should be built at least according to Grade II standard, using the Ministry of Communications scheme of highway classification. This implies that their design speed can reach 80 km/h (Ministry of Communications, 1994). But Grade II roads generally have only two lanes. Traffic can be held up easily due to the presence of a slow moving truck or frequent stopping of vehicles for loading and unloading. Many parts of the NHS are built according to Grade I standard. These roads are wider. But still they are generally not grade separated and are therefore subject to such factors as presence of traffic lights and frequent stopping. Considering the mix of traffic on China's roads, which is largely composed of slow moving heavy vehicles, tractors serving as goods vehicles and non-motorized traffic, and also the mountainous terrain in most parts of the country, an average speed of 40 km/h for the unimproved sections of the NHS is therefore a reasonable guess. As for motorways, the design speed is 120 km/h for light vehicles and 80–100 km/h for heavy vehicles. Thus the assumed average speed of 100 km/h again is a reasonable guess. [3]

Hainan Island presents some difficulty, as it is not connected to the mainland by road. It is assumed that ferry speed across Leizhou Strait, which separates Hainan from the mainland, averages 10 km/h, after taking into consideration vehicle waiting time.

With these assumptions about average traffic speeds, we can proceed to measure the changing time distances across the nation with the gradual implementation of the NTHS. The map of the proposed NTHS (Fig. 1) is first digitized. Physical distance for each highway segment in this network is given by the Geographical Information System (GIS) package, Arc/View. The corresponding time distance can then be obtained by dividing the physical distance by the assumed average highway speed for each of the following years, depending on whether the section has been upgraded to motorway or not:

(i) 1990, base year or the year when implementation of the NTHS began;
(ii) 1997, the year when the latest information on motorways opened to traffic is available;
(iii) 2000, the year when NTHS will form a true network, with the completion of two vertical and two horizontal routes;
(iv) 2020, the year when the entire NTHS is due for completion.

The minimum time distance matrix can be obtained by using the formula $S = L^n$. Alternatively, it can be obtained directly by applying the relevant function in Arc/View.

3. Analytical results

Table 1 gives the list of row marginals of the time distance matrix, a_i, the minimum time needed to travel from city i to all other cities, for each of the four years under study. Also given are $\sum a_i$'s and the associated coefficients of variation (CV), and the absolute and percentage change of the accessibility scores over the various study periods. Figs. 2(a) and (c) map the contours of accessibility for 1990, the base year, 2000, the year when the NTHS forms a true network, and 2020, the completion date of the project. Figs. 3(a) and (b) show the regional variation in accessibility improvement

[3] Alternative assumptions about travel speeds will, of course, yield different time distances. However, the emphasis here is on travel time reductions. Provided that speeds within the NTHS are substantially higher than those in sections of the NHS outside the NTHS, which is certainly the case, the reductions in travel times, both in absolute and relative terms, for the individual cities in the system are unlikely to exhibit substantial differences under alternative specifications of travel speeds.

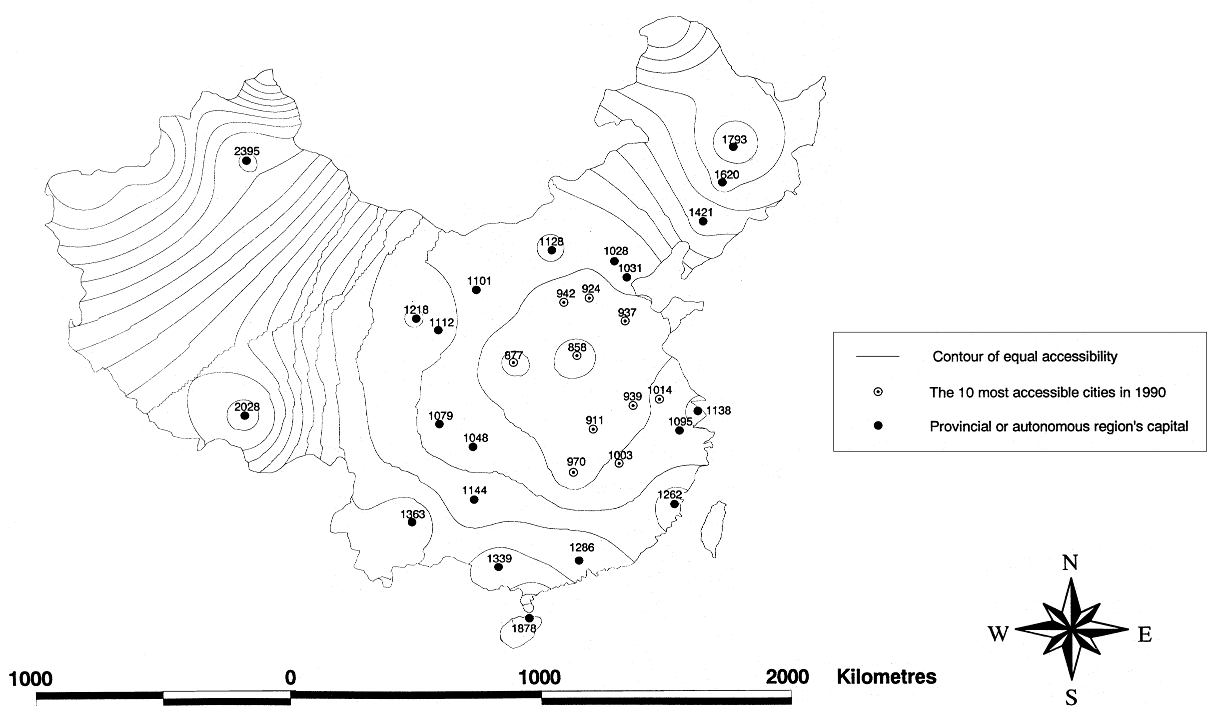
Table 1
China's NTHS: nodal accessibility 1990–2020

City	Accessibility score (h)				Change in accessibility			
	Year				Period			
	1990	1997	2000	2020	90-00		00-20	
					Hours	%	Hours	%
Beijing	1028.3	905.6	622.6	526.3	405.7	39.5	96.3	15.5
Tianjin	1030.7	918.7	636.7	525.5	394.0	38.2	111.2	17.5
Shijiazhuang	924.4	850.6	572.1	475.5	352.3	38.1	96.6	16.9
Taiyuan	941.7	862.4	622.2	505.6	319.5	33.9	116.6	18.7
Hohhot	1127.5	1032.3	800.0	621.0	327.5	29.0	179.0	22.4
Shenyang	1420.6	1321.3	787.1	682.7	633.5	44.6	104.4	13.3
Changchun	1619.5	1544.0	1012.4	899.3	607.1	37.5	113.1	11.2
Harbin	1793.1	1729.6	1087.6	973.2	705.5	39.3	114.4	10.5
Shanghai	1135.6	977.6	627.1	537.5	508.5	44.8	89.6	14.3
Nanjing	1014.0	900.6	674.0	525.2	340.0	33.5	148.8	22.1
Hangzhou	1095.4	1048.7	672.0	557.9	423.4	38.7	114.1	17.0
Hefei	939.3	861.7	662.9	474.4	276.4	29.4	188.5	28.4
Fuzhou	1262.2	1177.9	722.5	618.4	539.7	42.8	104.1	14.4
Nanchang	1003.4	921.5	715.9	523.6	287.5	28.7	192.3	26.9
Jinan	937.4	864.5	626.7	482.0	310.7	33.1	144.7	23.1
Zhengzhou	858.0	782.9	520.4	445.5	337.6	39.3	74.9	14.4
Wuhan	910.9	862.9	658.1	574.7	252.8	27.8	83.4	12.7
Changsha	996.8	956.7	693.6	564.4	303.2	30.4	129.2	18.6
Guangzhou	1285.6	1205.5	815.0	755.4	470.6	36.6	59.6	7.3
Nanning	1339.1	1243.7	827.8	703.8	511.3	38.2	124.0	15.0
Haikou	1678.4	1542.2	1005.6	896.8	672.8	40.1	108.8	10.8
Chengdu	1078.6	957.9	762.5	557.8	316.1	29.3	204.7	26.8
Chongqing	1047.8	994.0	726.4	573.1	321.4	30.7	153.3	21.1
Guiyang	1144.0	1074.1	770.8	590.5	373.2	32.6	180.3	23.4
Kunming	1363.2	1235.5	953.9	653.2	409.3	30.0	300.7	31.5
Lhasa	2028.3	1922.4	1731.9	1275.4	296.4	14.6	456.5	26.4
Xian	877.3	820.3	550.7	469.4	326.6	37.2	81.3	14.8
Lanzhou	1111.7	1053.9	681.0	619.2	430.7	38.7	61.8	9.1
Xining	1218.3	1145.5	793.1	716.9	425.2	34.9	76.2	9.6
Yinchuan	1101.3	1018.6	852.0	682.2	249.3	22.6	169.8	19.9
Urumqi	2395.0	2316.2	1207.0	1146.0	1188.0	49.6	61.0	5.1
Sum	37707.0	35030.6	24401.5	20152.2	13305.5	35.3	4249.3	17.4
Mean	1216.39	1130.62	786.83	650.08	429.54		136.75	
CV	0.293	0.311	0.301	0.306				

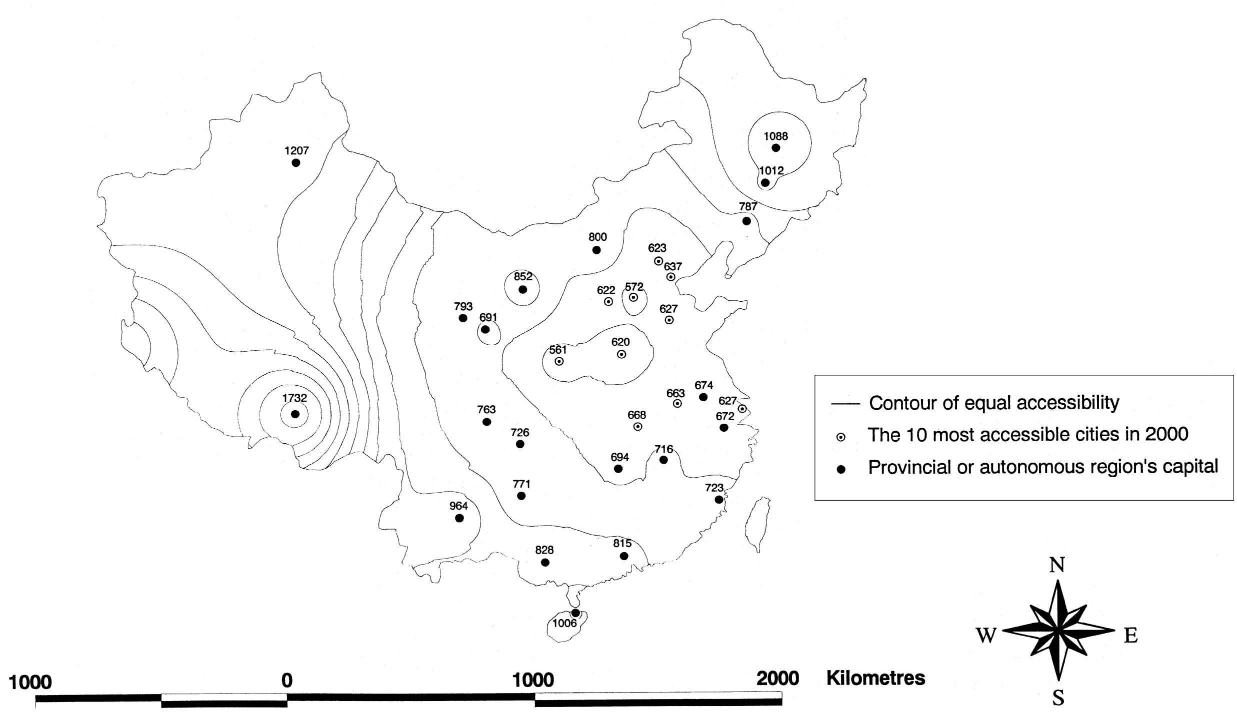
for the periods 1990–2000 and 2000–2020, respectively. A few points are evident from these calculations and mapping exercises. First, not surprisingly, road accessibility in China as a whole shows significant improvements with the gradual implementation of the NTHS. But what is more revealing from the results is that the first phase of the NTHS, to be completed in 2000, seems to have the greatest effects on accessibility improvement. $\sum a_i$, the minimum time needed to travel from all cities to all other cities in the network, decreased from 37,707 h in 1990 to 35,031 h in 1997, a drop of 2764 h or 7%. This sum decreases further to 24,402 h in 2000, when the two vertical routes and the two horizontal routes are completed. A major reason for this remarkable decline in total travel time–a drop of 10,629 h or 30% over a period of only three years–is that highway construction takes time. Most sections due for completion in 2000 would have commenced construction in the mid 1990s. Another reason, which is perhaps more fundamental, is that with the completion of this first phase of NTHS, a true network in the graph theoretical sense will be formed.

The second phase of the project, which will be implemented in the period 2000–2020, involves more kilometres of road than the first phase. Also, the associated highway segments are mainly found in the western and southwestern parts of the country where the geologic and geomorphic conditions are more difficult for highway construction. Thus, the second phase will likely be much more costly as well. However, its effects on reducing time distance do not appear to be as significant as those of the former phase. Further addition of motorway segments to the network no doubt will increase connectivity and hence network accessibility. But its effects are likely to be less significant once the basic structure of the network has been formed. Thus, the results show that in the year 2020, when the full network is completed, $\sum a_i$ will only decrease to 20,152 h, or a mere 4249 h below the 2000 figure. This may be compared with the total reduction of 13,305 h for the first phase. Considering the geographical locations of the new additions to the motorway system and their cost implications, it is not difficult to arrive at the following conclusion. The benefit/cost ratios for these new addi-

(a) 1990



(b) 2000



(c) 2020

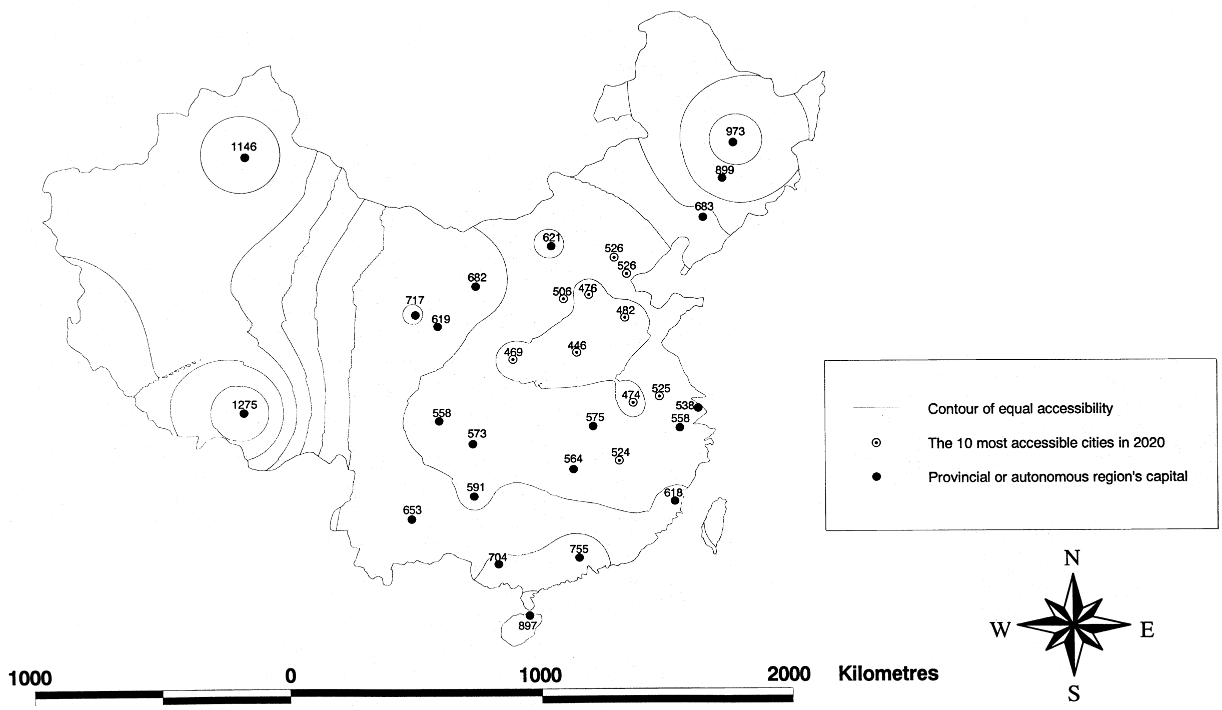


Fig. 2. The NTHS: travel time accessibility: (a) 1990; (b) 2000; (c) 2020.

tions are even less favourable than what the decreasing levels of time reductions would have suggested. Highway investments, like investments in other sectors of the economy, may well exhibit diminishing returns.

Second, Zhenzhou, provincial capital of Henan, constantly ranks highest in terms of the nodal accessibility scores (i.e., the lowest travel time distance to all other places, a_i). This is to be expected given that most provincial capitals are located in the eastern half of the country and Zhengzhou is located almost at the geometric centre of China's eastern half. The contours of accessibility gradient broadly correspond to the physical geography and population distribution of China. Other cities located in the central and east-central part of the country such as Shijiazhuang, Xian and Wuhan also rank high in the accessibility score. Again, what is more revealing is the changes in the accessibility score, as measured by a_i, exhibited by individual cities over time. A major concern of the Chinese planners is the increasing disparity between the coastal provinces and the country's vast interior hinterlands (see, for example, Li and Tang, 2000). It is therefore useful to examine whether the NTHS has differential effects on different parts of the country. In particular, it is important to assess whether the effects on the more prosperous coastal areas and on the lagging interior, especially the remote southwestern provinces, differ significantly. For this we examine the maps showing the change in accessibility (Figs. 3(a) and (b)).

To reiterate, the greatest reductions in travel times occur in the first phase (1990–2000) of the project. The mean time reduction for the 31 urban centres for this period is 429.5 h. But for the second period (2000–2020) it is only 136.7 h. Thus the regional impacts are mostly felt in the first period. Improvements in the second period are more of an incremental nature. Urumqi stands out to be the city that experiences the largest improvement in accessibility in the first phase, whether measured in absolute (1188 h) or percentage (49.6%) terms. But the improvements recorded for other western cities are much smaller. Cities such as Kunming, Chengdu, Chongqing, Guiyang in the southwest where the terrain is particularly rugged all experience less than average time reductions. Other interior cities such as Hohhot, Hefei, Nanchang and Changsha also show below average time reductions.

In contrast, cities in the coastal region generally report above average reductions in aggregate time distance in the first phase. Shanghai, China's foremost industrial and financial centre, reports a reduction of 509 h or 44.8% of the base year figure. Beijing and Tianjin show a reduction of 406 h or 39.5% and 394 h or 38.2%, respectively. It is widely recognized that China's most dynamic and rapidly growing province in the reform period is Guangdong, the capital city of which is Guangzhou. In percentage terms, Guangzhou experiences only slightly above average reduction in aggregate time distance. However, in absolute terms Guangzhou's position with

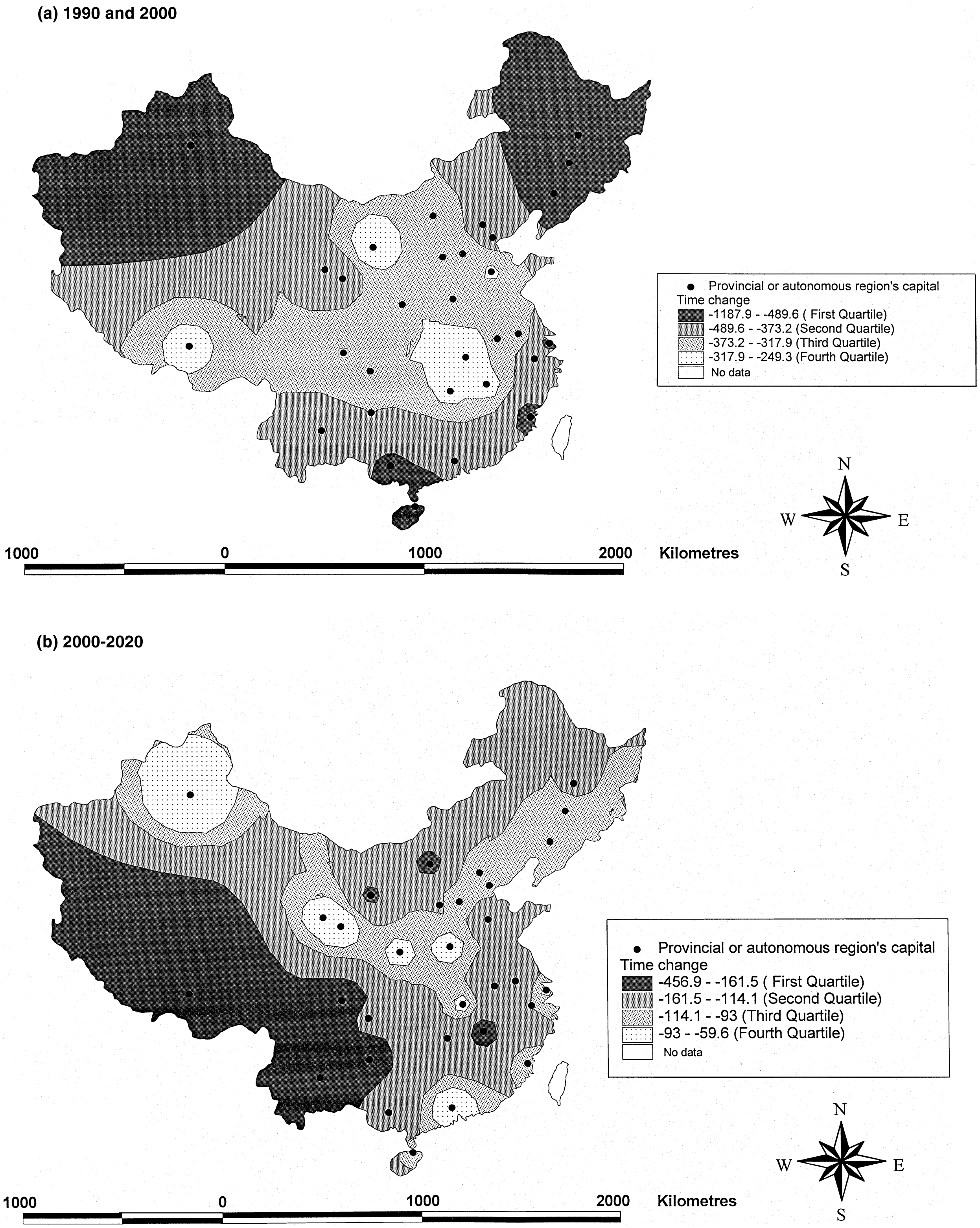


Fig. 3. Change in travel time accessibility: (a)1990 and 2000, (b) 2000–2020.

respect to accessibility has improved quite substantially. Another major beneficiary of China's economic reforms is Fujian Province, which is located on the southeastern coast facing Taiwan. Fuzhou, capital of Fujian, benefits greatly from the first phase of NTHS. The aggregate travel time recorded decreases by 540 h or 42.8%.

The region that benefits most from the first phase of NTHS appears to be the northeast. Almost all cities in this region report remarkable improvements in time distance. It may be noted that the northeast is the heartland of China's large-scale state-owned industrial enterprises. Most of these are suffering from antiquated machinery and mismanagement, and are on the brink of bankruptcy. In fact the term "northeast syndrome" has been coined to describe the rapid decline in the economic strength of the northeast since the launching of the market-oriented reforms (Li, 2000). Most measures of regional disparity in China exhibit a rapid declining trend at least in the first 10–15 years of the reform (Li, 2000). A large part of it has to do with the relative and probably even absolute decline of the northeast. The much improved accessibility positions of the northeast as a result of the implementation of the NTHS may help halt the region's decline.

From the above, it may be concluded that the first phase of NTHS in general tends to aggravate China's east versus west or coastal versus inland divide, although this statement has to be qualified by the remarkable improvement shown by Urumqi and cities in the northeast. In the latter case, the cities and their respective provinces are still considered part of the "east", but definitely they are the losers in the market-oriented reforms. Thus, the coefficient of variation computed from the accessibility scores only increases slightly from 0.29 in 1990 to 0.31 in 1997, and then slips back to 0.30 in 2000. But the positions of the lagging southwestern provinces certainly become worse after the completion of the first phase of the project.

This aggravation of regional disparity is somewhat mitigated in the second phase when the motorway system further extends to the southwest and other interior provinces. The coefficient of variation of the accessibility scores stays at more or less the same level. However, an examination of Table 1 and the figures shows that, with the completion of the entire network in 2020, the distribution of the accessibility scores will become spatially more uniform, and the accessibility gradients will become much gentler. The largest improvements in the second phase are found in the southwest and other interior provinces. Cities such as Chengdu, Guiyang, Kunming, Hohhot, Hefei and Nanchang all experience more than 23% or more of aggregate travel time reduction over the period, as compared with the national average of 17.4%. Lhasa, capital of Tibet, in particular, reports a 457 h reduction in aggregate travel time, which is the largest reduction recorded for this phase. On the other hand, cities in the eastern coastal region such as Beijing, Shanghai and Fuzhou generally report below average reductions in aggregate travel time in this period. Cities in the northeast and the northwest, which benefit substantially from the NTHS in the first phase, in general also report below average aggregate travel time reductions in the second phase.

4. Concluding remarks

The changing accessibility gradients over time resulting from the phased implementation of the NTHS to some degree confirms Williamson (1965) "inverted U" hypothesis. The computations show that the NTHS tends to bring about worsening spatial disparity in its first phase of development, especially if attention is given to the lagging and mountainous southwest. However, as time progresses and as the NTHS extends to the more remote parts of the country, the NTHS is likely to result in spatially more balanced growth. The fact that the first phase of the NTHS focuses on the more prosperous and more densely populated coastal provinces is understandable. Motorway construction is extremely costly and is economically justifiable only if there exist substantial amounts of traffic demand. It has been argued that in most parts of the world, dirt roads with only a single lane are all that is needed (Thomson, 1974, p. 8). Given China's current income levels, motorways may not be economically viable even in the most prosperous regions of the country. In the more remote and lagging parts of China where traffic is light, the squeezing effects caused by massive investments in motorway construction could be acute.

However, the local governments in Sichuan, Guizhou and other remotely located provinces would argue that only through substantial investments in transport infrastructures can the adversities due to their disadvantageous geographical positions be mitigated. Recently the Chinese government has pronounced a new regional development policy, the so-called "grand development of China's west (*xibu dakaifa*)" policy. Measures have been introduced to facilitate the transfer of resources to the western interior. Development of highway and other transport infrastructures are top on the priority. It is a general belief in China these days that transport investments should take priority over other investments. The mass media are flooded with sayings such as *lutong caitong*, or wealth will follow extension of roadways, preferably motorways. Whether transport infrastructures should follow economic development or whether investment in transport infrastructure will act as a stimulator of economic development has been a topic of debate for many years (Hoyle and Smith, 1992). Apparently most local governments in China today opt to believe in the latter. Somehow the national government has to address the problem of spatial imbalance. Thus, even though the first phase of the NTHS in general favours the coastal region, extensive stretches of motorways in this phase are in fact located in the remote provinces in the southwest and northwest. Had the NTHS been implemented strictly on a cost-effective basis, its effects on worsening the country's already highly unbalanced pattern of spatial development, at least in the initial stages of the programme, would have been many times larger.

Earlier it was pointed out that railways until recently have formed the backbone of China's land transport. The construction of motorways, especially at the scale of the NTHS, is a recent phenomenon. Motorways and railways share many characteristics in common. These include safety, speed and volume (White and Senior, 1983, pp. 71–79). But motorways, in conjunction with networks of secondary and tertiary roads, also provide flexibility and door-to-door service. Whereas in a railway-based economy productive activities tend to concentrate on a selected number of nodes, i.e., the major rail stations and their vicinity, in a highway-based economy the momentum of growth is likely to be spatially diffuse. At the national level, the first stage of the NTHS may further aggravate spatial disparity in China. However, at the regional level, implementation of the NTHS at all stages is likely to bring about trickling down effects. But motorways also consume large quantities of valuable farmland and induce urban sprawl. Also, the motorcar as a mode of transport is notorious for its energy inefficiency and for its adverse effects on the environment. The current emphasis on motorway development is somehow at odds with China's quest for sustainable development, although it fits well with her desire to develop a thriving automotive industry (Wang and Weidong, 2000).

Acknowledgements

The authors would like to thank the anonymous referees for their valuable suggestions and Miss Doris Fung for computation and cartographic assistance. Of course, the remaining errors are ours.

References

Asian Development Bank. 1998. Report 1998, No. 30082 PRC. Asian Development Bank, Manila.

Bruinsma, F., Rietveld, P., 1998. The accessibility of European cities: theoretical framework and comparison of approaches. Environment and Planning A 30 (3), 499–521.

China Transport and Communications Yearbook, 1995. Yearbook Publication House of China Transport and Communications, Beijing.

Gabriel, D., Vaclav, S., 1996. Cities and highway networks in Europe. Journal of Transport Geography 4 (2), 107–121.

Gao, R., 1998. Kua Shiji Gonglu Jiaotong Fazhan Zhanlue (Strategy for Highway Development in the Turn of the Century). Renmin Jiaotong Chubanshe, Beijing.

Garrison, W.L., 1960. Connectivity of the Interstate Highway System. Papers and Proceedings of the Regional Science Association 6, 121–137.

Gutiérrez, J., Gómez, G., 1999. The impact of orbital motorways on intra-metropolitan accessibility: the case of Madrid's M-40. Journal of Transport Geography 7, 1–15.

Gutiérrez, J., Urbano, P., 1996. Accessibility in the European Union: the impact of the trans-European road network. Journal of Transport Geography 4 (1), 15–25.

Gutiérrez, J., Rafael, G., Gómez, G., 1996. The European high-speed train network. Journal of Transport Geography 4 (4), 227–238.

Hoyle, B.S., Smith, J., 1992. Transport and Development. In; Brian, S., Hoyle, R.D., Knowles, (Eds.), Modern Transport Geography. Wiley, Chichester, pp. 11–32.

Ivy, R.L., 1995. The restructuring of air transport linkages in the New Europe. Professional Geographer 47 (3), 280–288.

Li, S., 2000. China's changing spatial disparities, a review of empirical evidence. In: Li, S., Tang, W. (Eds.), China's Regions, Polity and Economy: A Study of Spatial Transformation in the Post-Reform Era. Chinese University Press, Hong Kong, pp. 155–186.

Li, S., Tang, W. (Eds.), 2000. China's Regions, Polity and Economy: A Study of Spatial Transformation in the Post-Reform Era. Chinese University Press, Hong Kong.

Linneker, B., Spence, N., 1996. Road transport infrastructure and regional economic development, the regional development effects of the M25 London orbital motorway. Journal of Transport Geography 4 (2), 77–92.

Loo, B.P.Y., 1999. Development of a regional transport infrastructure, some lessions from the Zhujinag Delta, Guangdong, China. Journal of Transport Geography 7 (1), 43–63.

Ministry of Communications, PRC, 1994. Ministerial Standards of the Ministry of Communications of the People's Republic of China, Highways Engineering Standard JIJ 01-89. New World Press, Beijing.

Murayama, Y., 1994. The impact of railways on accessibility in the Japanese urban system. Journal of Transport Geography 2 (2), 87–100.

State Statistical Bureau (SSB), 1998. China Statistical Yearbook 1998. Tongji Chubanshe, Beijing.

Taaffe, E.J., Gauthier., H.L., 1973. Geography of Transportation, 1st ed. Prentice-Hall, Upper Saddle River, New Jersey.

Taaffe, E.J., Gauthier, H.L., O'Kelly M.E., 1996. Geography of Transportation. 2nd ed. Prentice-Hall, Upper Saddle River, New Jersey.

Tang, W.-S., 1997. Urbanization in China: A review of its causal mechanisms and spatial relations. Progress in Planning 48 (1), 1–65.

Thomson, J.M., 1974. Modern Transport Economics. Penguin, Middlesex, England.

Wang, J.J., Liu, W., 2000. The dynamic relationship between the central government and local authorities in orienting foreign direct investment, a case study of the automotive industry. In: Li, S., Tang, W. (Eds.), China's Regions, Polity and Economy: A Study of Spatial Transformation in the Post-Reform Era. Chinese University Press, Hong Kong, pp. 217–246.

Wheeler, D.C., O'Kelly, M.E., 1999. Network topology and city accessibility of the commercial interest. Professional Geographer 51(3), 327–339.

White, H.P., Senior, M.L., 1983. Transport Geography. Longman, London.

Williamson, J.G., 1965. Regional inequality and the process of national development, a description of the patterns. Economic Development and Cultural Change 13, 3–45.

Zhang, L., Zhao, S.X.B., 1998. Re-examining China's urban concept and the level of urbanization. The China Quarterly 154, 330–381.